



CT117-3-2-FWDD Further Web Design and Development

## Individual Assignment Proposal

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|---------------|----------------------------------------------------|
| Name          | Taudea Arehui Gabi                                 |
| Student ID    | TP083304                                           |
| Intake        | APD2F2509SE                                        |
| Project Title | Python Adventure Quest: A Hybrid Educational Game. |

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| The chosen Python programming concept                  | Control Flow                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| The role of the Web App in the educational hybrid game | <ul style="list-style-type: none"> <li>▪ Allow players to input answers to questions during the game.</li> <li>▪ Automatically check answers and provide feedback.</li> <li>▪ Track player scores and progress throughout the game.</li> <li>▪ Display hints and explanations when players answer incorrectly.</li> <li>▪ Show leaderboard or final results at the end of the game.</li> </ul>                                                                                                                                                                                                                                                                                                                                     |
| The physical component of the game                     | <p>Physical Component(s):</p> <ul style="list-style-type: none"> <li>▪ <b>Game Board</b><br/>A printed board that represents a path or adventure map. Players move along the board as they answer questions correctly.</li> <li>▪ <b>Player Tokens</b><br/>Small pieces representing each player. Used to track position on the board.</li> <li>▪ <b>Question Cards</b><br/>Cards containing Python control flow questions. Players draw a card when they land on certain spaces.</li> <li>▪ <b>Event Cards</b><br/>Cards that create special events (move forward, move backward, skip turn). Adds excitement and randomness to the game.</li> <li>▪ <b>Dice</b><br/>Used to determine how many spaces a player moves.</li> </ul> |
| Specifications for the end user                        | <p><b>Student (Main User)</b></p> <ul style="list-style-type: none"> <li>▪ Able to access the web app through a browser.</li> <li>▪ Can view programming questions on the web interface.</li> <li>▪ Can input answers to Python questions.</li> <li>▪ Receive instant feedback (correct or incorrect).</li> <li>▪ Earn points for correct answers.</li> <li>▪ Track their progress and score throughout the game.</li> </ul> <p><b>MC</b></p> <ul style="list-style-type: none"> <li>▪ Can start or reset the game.</li> <li>▪ Monitor player scores through the web app.</li> <li>▪ Guide players on how to play the hybrid game.</li> </ul>                                                                                      |

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|  | <ul style="list-style-type: none"><li>▪ Assist players when they need help understanding questions.</li></ul> |
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